Set associativity

Tuesday, May 11, 2021 9:39 AM

Example 2 – SA caches

$MM_{size} = 4\,GiB = 2^2 \cdot 2^{30}\,B = 2^{32}\,B$

- Byte – addressable
- 1 word = 4 B => 2^2 B
- 1 block = 32 words = 2^5 words = $2^5 \cdot 2^2\,B = 2^7\,B$
- 4 way SA
- Cache data size = 256 KiB

a) • Memory address format

b) • Cache diagram

$\frac{256\,KiB}{2^7\,B} = \frac{2^{18}\,B}{2^7\,B}$ Blocks/Cache $= 2^{11}$

$\frac{2^{11}}{2^2} = 2^9$ Blocks/Bank

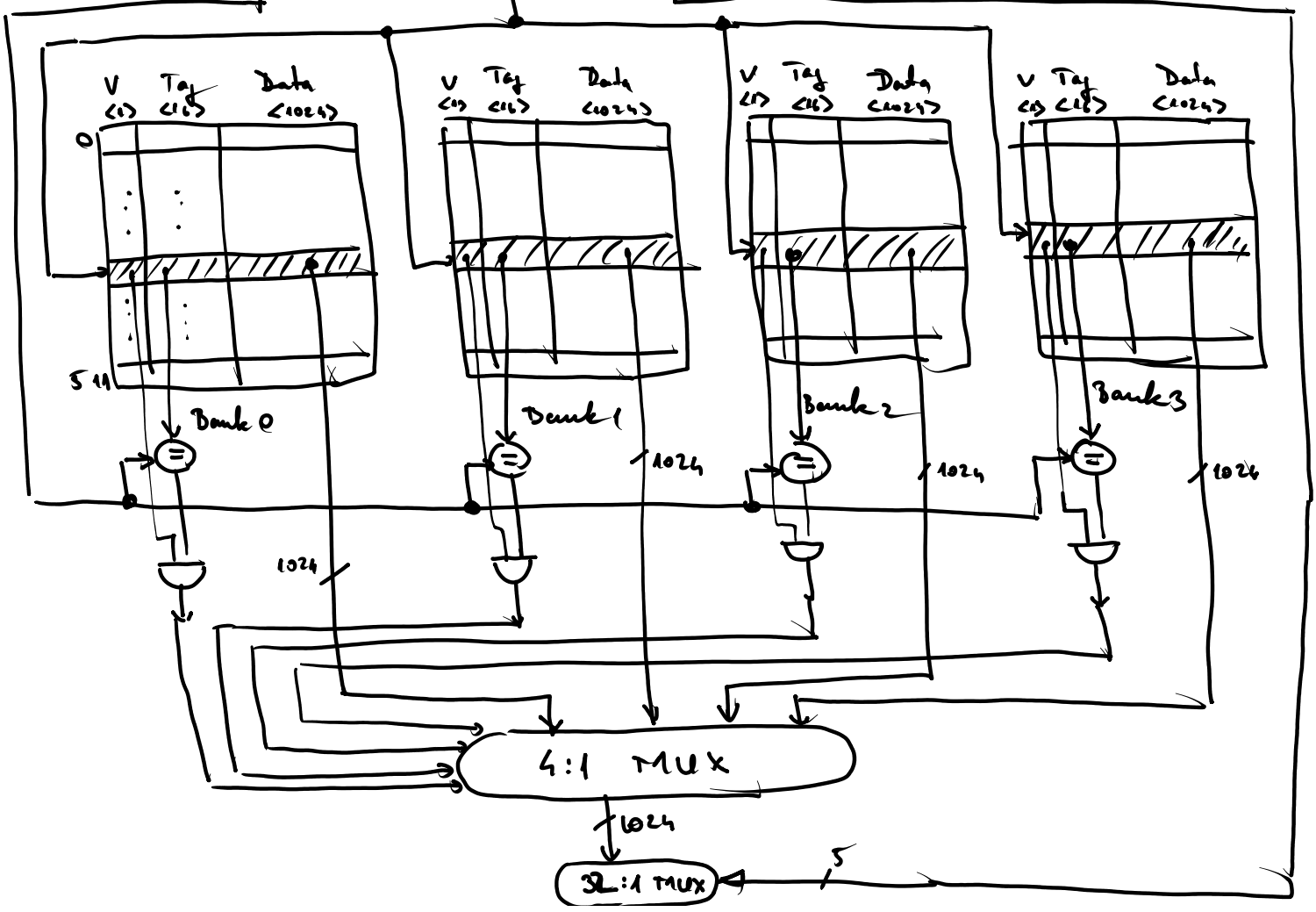
a)

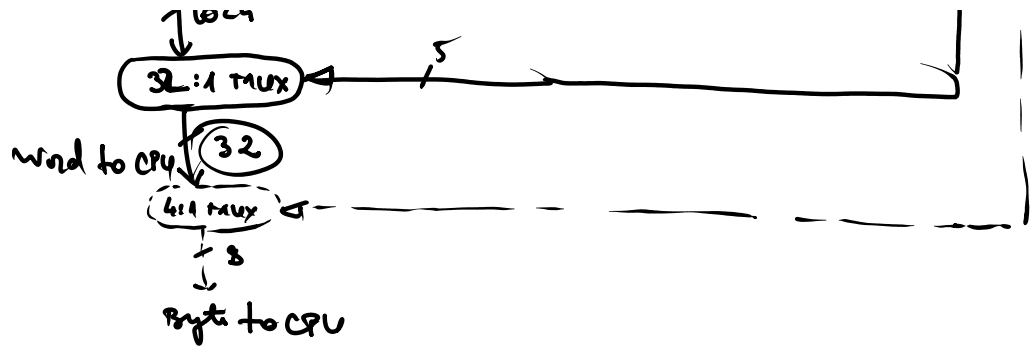
31 … 16	15 … 7	6 5 4 3 2	1 0
Tag	Index	Block offset	word offset
<16>	<9>	<5>	<2>

b)

<16>	<9>	<5>	<2>
Tag	Index	Block offset	word offset

Address word



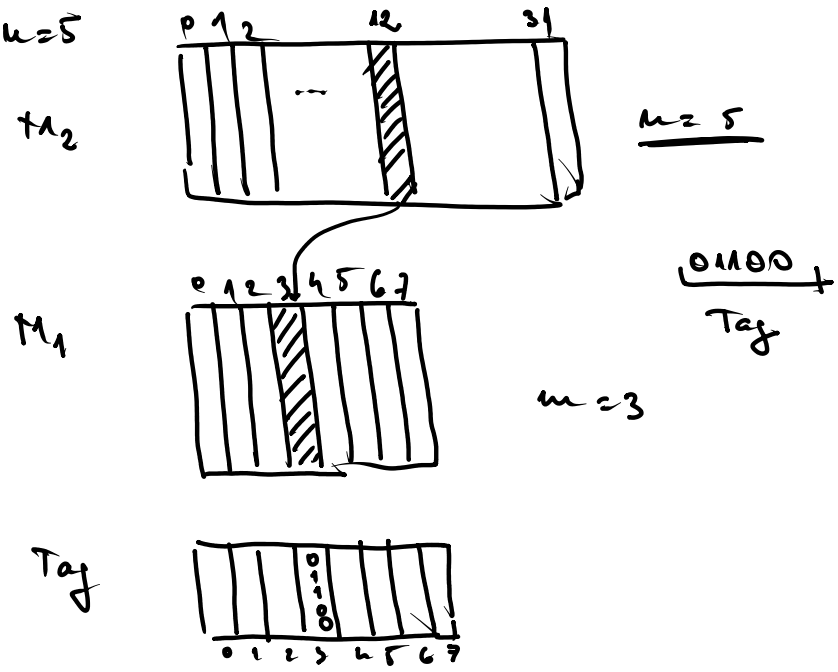


3.2.3 Full-associative caches

$M_2 \rightarrow 2^n$ blocks : $M_2(0) \dots M_2(2^n-1)$

Cache $M_1 \rightarrow 2^m$ blocks : $M_1(0) \dots M_1(i) \dots M_1(2^m-1)$

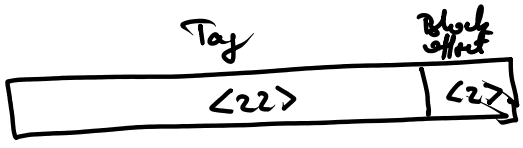
Example $m=3$, $n=5$



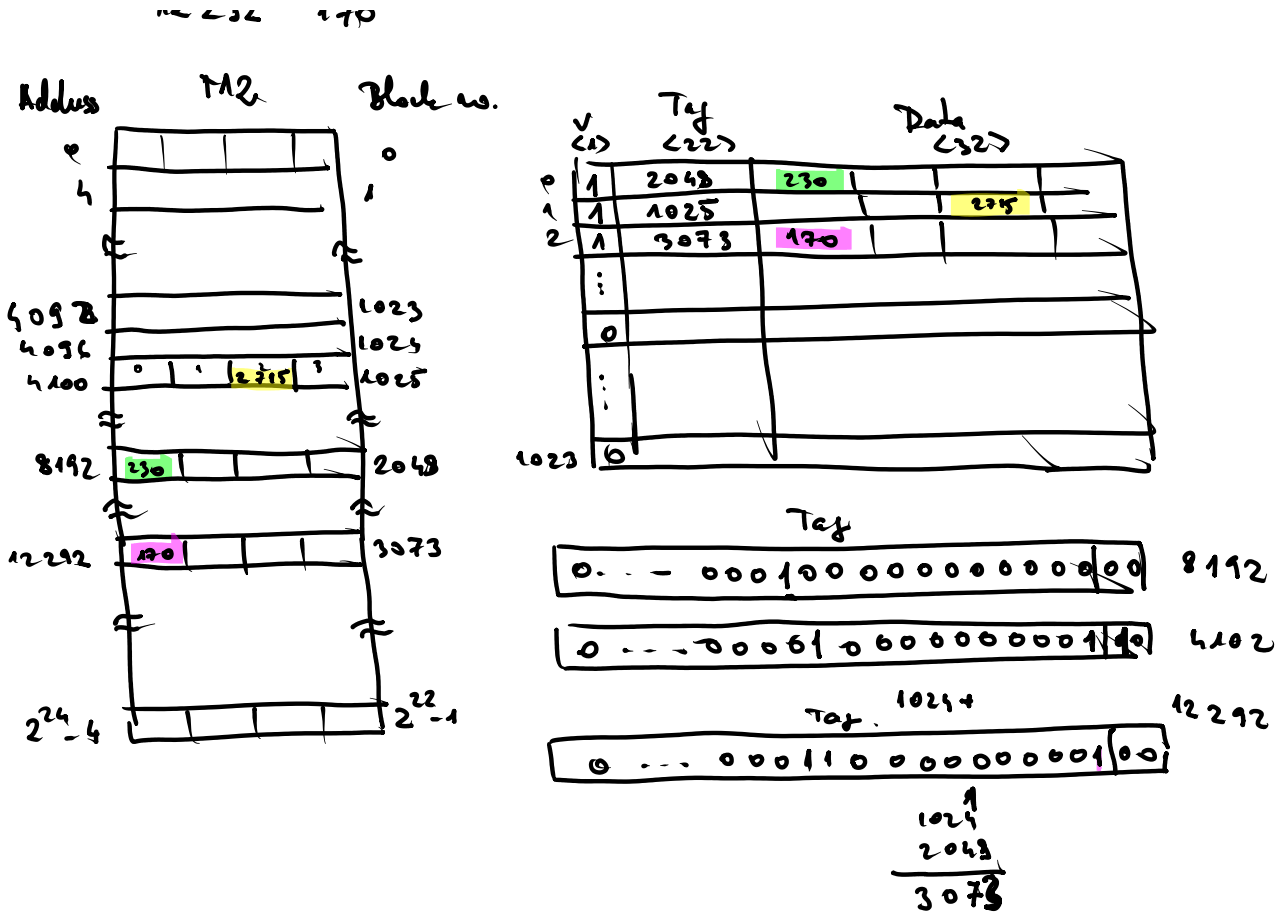
FA Solution
- k-way SA ($k=2^s$) $k=2^m$, $\boxed{s=m}$
- The most compliant with the locality principle
- The problem → increase the t_{ac} ⇒ reduced miss rate

Example

Addr	Code
8192	230
4102	2715
12292	170



Address M2 Block no.



3.3 Cache replacing methods

- Only for SA and FA caches
- Cache is full

3 Policies

a) Random : not compliant w/ the spatial locality principle (works for small SA)

b) LRU – Least Recently Used – the most compliant w/ the locality principle (temporal)

c) FIFO – first-in, first-out (queue) : not compliant with the locality principle

a) Random

Galois field polinomials

$G(x) = x^4 + x + 1$ Feedback shift register

rnd (G(x)) = k => seq. length $2^k - 1$